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# The Water Diffusion of Brain Following Hypoglycemia in Rats – A Study with Diffusion Weighted Imaging and Neuropathologic Analysis

**Fang Tong,[a†] Yijuan Zou,[bc†] Yue Liang,[a†] Hao Lei,[bc] Tenzin Lopsong,[a] Yuluo Liu,[a] Jehane Michael Le Grange,[d] Guanglong He[e*] and Yiwu Zhou[a*]**

[a]*Department of Forensic Medicine, Huazhong University of Science and Technology, Tongji Medical College, No. 13 Hangkong Road, Hankou, Wuhan, 430030, PR China*

[b]*Wuhan National Laboratory for Optoelectronics, Huazhong University of Science and Technology, Wuhan 430074, PR China*

[c]*National Center for Magnetic Resonance in Wuhan, State Key Laboratory of Magnetic Resonance and Atomic and Molecular Physics, Wuhan Institute of Physics and Mathematics, Chinese Academy of Sciences*

[d]*The First Clinical college, Wuhan Union Hospital, Tongji Medical College, Huazhong University of Science and Technology, 1277 Jiefang Avenue, Wuhan, 430022, PR China*

[e]*Institute of Forensic Science, Ministry of Public Security, People's Republic of China, No. 17 Nanli Mulidi, Beijing 100038, China*

**Abstract—We established hypoglycemic rat models and divided them into three groups (the sham group, the acute hypoglycemia group and the recovery group). The brain water diffusion was examined using DWI. Thereafter, neuropathologic examinations were performed in order to evaluate the distribution of brain damage. The expression of AQP4 and Caspase3 was also examined using Western blot. We aimed to determine the specific brain regions which were vulnerable to hypoglycemia in relation to the water diffusion and neuropathology. The DWI scanning showed abnormal water diffusion in the cortex, hippocampus and hypothalamus during each stage of hypoglycemia. In the acute hypoglycemia group, the apparent diffusion coefficient (ADC) of the dentate gyrus (DG) and the hypothalamus was increased, while the ADC of the somatosensory cortex (SSc), subcortex and striatum (Str) was decreased. After glucose reperfusion and a 7-day recovery period, most of the hypoglycemia-induced changes in ADC returned to normal, except in the hypothalamus, posterior SSc and DG, which demonstrated increased ADC levels. The lowest AQP4 expression was observed in the cortex of the acute hypoglycemia group. Furthermore, there was increased Caspase3 expression in the hippocampus of the recovery group. The expression of Caspase3 in the hypothalamus was most prominent in the acute hypoglycemia group. Our work revealed that hypoglycemia significantly influenced the water diffusion of the brain. The decrease of AQP4 was associated with the formation of cytotoxic edema in acute hypoglycemia. Hypoglycemia primarily tends to damage the cerebral cortex, hippocampus and hypothalamus and may result in permanent injury to the brain.** 

**Key words: DWI, hypoglycemia, neuropathology, water diffusion.**

## INTRODUCTION

Insulin is commonly used to treat diabetes, while its side effects have been questioned and are controversial (Auer, 1986). It has been reported that the improper use of insulin could induce severe hypoglycemia and may give rise to coma, seizure and even death (Auer et al., 1989; Del Campo et al., 2009). In addition, insulin could serve as the perfect murder weapon due to its ease of access and elusive pathologic changes (Auer et al., 1984; Marks and Wark, 2013; Tong et al., 2017).

Previous studies have discovered that brain damage induced by hypoglycemia is different from ischemic damage, especially in the hippocampal dentate gyrus (DG) which shows relative resistance to ischemia (Auer et al., 1985b, 1989; Auer, 2004). Ultrastructural studies of the hippocampus have demonstrated that hypoglycemia can leave vacuoles in place of perikarya undergoing dissolution and mild dilatation of the endoplasmic reticulum during different periods (Auer et al., 1985b). The electroencephalogram (EEG) is used to determine the presence of hypoglycemic brain damage, and hypoglycemia-induced neuronal death in the hippocampus appears when the EEG becomes flat (Auer, 2004). According to previous studies, hypoglycemia could induce two kinds of brain edema, namely cytotoxic

*Corresponding authors. Tel.: +86 27 83657925; fax: +86 27 83692638, +86 13707160935 (cell).
E-mail address: leihao@wipm.ac.cn (Hao Lei).guanglong_he@163.com (Guanglong He).zhouyiwu@outlook.com (Yiwu Zhou).
† Fang Tong, Yijuan Zou and Yue Liang contributed equally.

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and vasogenic types (Deng et al., 2014). Vasogenic edema stems from the damage of the BBB (brain blood barrier) and may result in increased water diffusion, while cytotoxic edema occurs only in the parenchyma without damage to the BBB and may decrease water diffusion (Klatzo, 1967; Gisselsson et al., 1998). Research has demonstrated that the expression of Aquaporin 4 (AQP4) changes when cytotoxic edema occurs. AQP4, one of the predominant water channels in the CNS, is found to exist on the end foot of the astrocyte and ependyma (Nielsen et al., 1997; Assentoft et al., 2015). The brain edema induced by ischemia or intoxication is associated with the expression of AQP4 (Manley et al., 2000; Papadopoulos and Verkman, 2005; Assentoft et al., 2015). Since edema is a conspicuous pathological change in hypoglycemia, we speculated that AQP4 plays an important role in this process.

DWI is thought to be sensitive to brain lesions induced by hypoglycemia and the apparent diffusion coefficient (ADC) value by DWI can mirror the status of water diffusion (Finelli, 2001; Lo et al., 2006). In the practice of clinical radiology (Johkura et al., 2012; Ohshita et al., 2015; Ren et al., 2017), DWI can be exploited to weigh the prognostic evaluation of hypoglycemic encephalopathy. Decreased water diffusion is highly suggestive of cytotoxic injury in most instances while it is increased in vasogenic edema (Mintorovitch et al., 1991; Latour et al., 1994; Schlaug et al., 1997). However, clinical case reports and trials have some limitations due to selection bias and other illnesses (e.g., ischemia, tumors) which cannot be entirely excluded. To eliminate these factors and perform a postmortem examination for further research, we designed the present study using animal models.

Cysteine-aspartic acid protease 3 (Caspase3) is a member of caspase family and cellular apoptosis is associated with the activation of the Caspase cascade (Gervais et al., 1999). Caspase 3 is pivotal in this process (Alnemri et al., 1996; Salvesen, 2002; Ghavami et al., 2009). Research has shown that Caspase 3 can convert β-Amyloid precursor protein (β-APP) into β-Amyloid (Aβ) which can lead to neuronal death (Gervais et al., 1999). Thus, the expression of Caspase 3 can be utilized to evaluate whether neuronal apoptosis is initiated.

This paper aims to determine the regional-specific vulnerability to severe hypoglycemia in terms of the water diffusion and neuropathologic changes. The canonical hypoglycemic animal model, where insulin-induced hypoglycemia with a flat-line EEG for 1 h is used, was applied in this research (Auer, 2004). Evaluation of the function of AQP4 in hypoglycemia-induced edema was also included in the present study. The results of this paper may provide additional knowledge on the neuropathologic changes of hypoglycemia and assist in its diagnosis in both clinical and forensic practice.

## EXPERIMENTAL PROCEDURE

### Animal groups and hypoglycemic animal model

Thirty male Sprague–Dawley rats weighing 250–300 g were used. The rats were randomly divided into 3 groups: the sham group, the acute hypoglycemia group and the recovery group (n = 10 for each group). All procedures were approved under the local guidelines for the care and use of animals. The hypoglycemic animal models were established according to the method previously reported (Auer et al., 1984, 1985b). All rats were fasted for 24 h before the experiment. The acute hypoglycemia group and the recovery group were injected with insulin (Novo 30R, Denmark) at 15 IU/kg body weight intraperitoneally (i.p.) and a hypoglycemic coma was induced. The standard chosen to confirm successful induction of the hypoglycemic model, was a continuous cerebral flat-line EEG for 1 h after insulin administration. In the acute hypoglycemia group, DWI scanning was performed immediately after a continuous, 1 h cerebral flat-line EEG was obtained. In the recovery group, after a successful hypoglycemic model was established according to the above criteria, a 50% glucose solution (i. p.) was slowly administered to the rats until their blood glucose rose to 3.9 mmol/L and this level was maintained. Thereafter, the recovery group was required to survive for 7 days before performing DWI scanning. After the DWI scanning, all rats were humanely sacrificed, and their brain tissues were collected for further study. Parts of the brain tissue (n = 5 for each group) were fixed in a 4% paraformaldehyde solution for pathologic analysis and the remaining tissues (n = 5 for each group) were homogenized for Western blot analysis.

### DWI scanning and data processing

The rats of each group were anesthetized with 1.8–2.5% isoflurane in pure oxygen to ensure stable respiratory and circulatory function during the DWI scanning. The rats were placed in a magnetic resonance-compatible stereotaxic frame of a 7.0-T/20 cm Bruker Biospec scanner (Karlsruhe, Germany) with a 72 mm-diameter volume coil for radio frequency transmission and a 40 mm-diameter quadrature surface coil for signal detection. Body temperature was maintained at 37 ± 1 °C by heated, circulating water, and respiration was maintained by a ventilator throughout the scanning. A spin-echo echo-planar DWI sequence was used with the following acquisition parameters: repeated time (TR) of 5000 ms, echo time (TE) of 28 ms, flip angle of 90°, slice thickness of 0.8 mm, slice interval of 0 mm, field of view of 30 × 30 $mm^2$ and matrix size of 128 × 128. The b value used for the acquisition of diffusion weighted images was 0, 200, 400, 600, 800, 1000 $s/mm^2$. The scanning area was set ranging from the frontal lobe (FL) to the occipital lobe (OL), and the total imaging time was approximately 8 min.

### H&E staining

The fixed brain tissues were embedded and sectioned into 5 μm sections. For H&E staining, the sections were dewaxed in xylene, rehydrated with ethanol solutions of decreasing concentrations, and washed in tap water. The sections were stained with hematoxylin for 5 min, washed with tap water, dehydrated with ethanol solutions of increasing concentrations, and stained with eosin for 5 min. Finally,

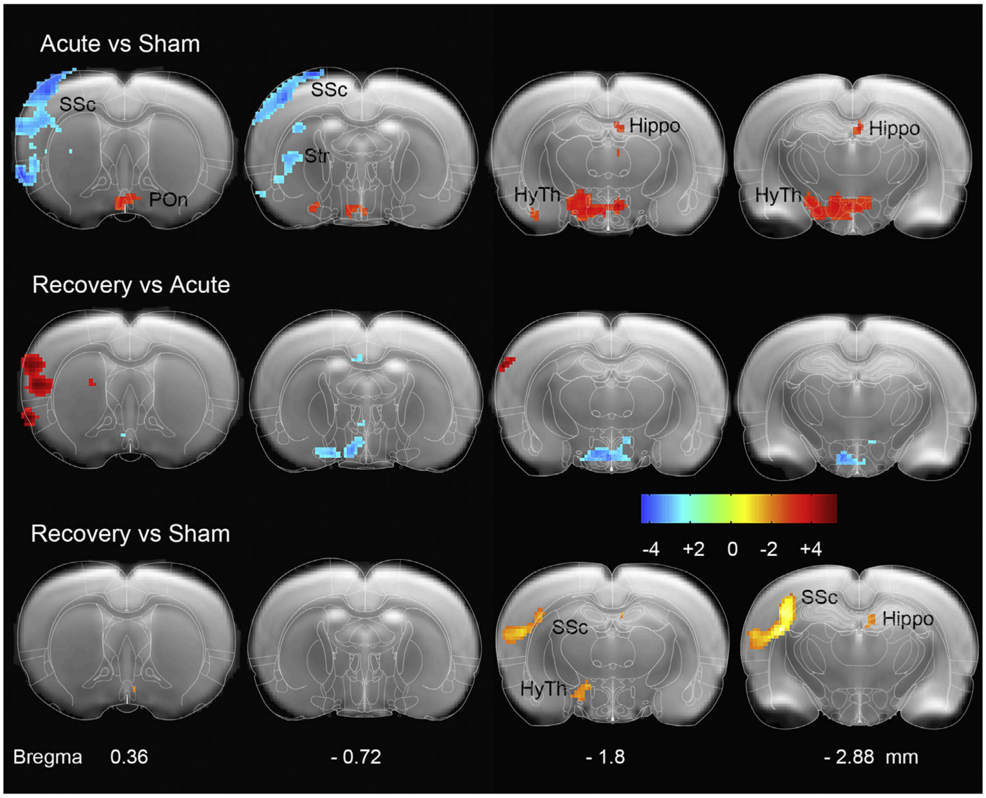


**Fig. 1.** Differential water diffusion map. Acute hypoglycemia group: Increased ADC of DG and hypothalamus (HyTh); decreased ADC of SSc, subcortex and Str. Recovery group: increased ADC in HyTh and DG compared to the sham group but decreased when compared to the acute hypoglycemia group; increased ADC in the posterior SSc and subcortical region. The bar demonstrated the general T-score of the ADC change.

the sections were dehydrated, hyalinized in xylene and mounted. All the sections were examined under a Nikon Eclipse 90i microscope (Tokyo, Japan). Digital pictures of H&E staining were measured using Image Pro Plus 6.0 (http://www.mediacy.com/) to evaluate the size of the neurons in each group.

## Silver staining

The fixed brain tissues were embedded and sectioned into 5 μm sections. For silver stain impregnation, the sections were de-waxed in xylene, rehydrated with ethanol solutions of decreasing concentrations, and washed in tap water. The sections were stained with $AgNO_3$ solution for 10 min and washed with tap water. Thereafter, sections were transferred into the developing liquid for at least 1 min and dried at 37 °C. Finally, the sections were dehydrated, hyalinized in xylene and mounted.

## Western blot analysis for AQP4 and Caspase3

Tissues of cerebral cortex (mainly containing SSc), hippocampus and hypothalamus were separated from the whole brain and individually homogenized in 0 °C RIPA lysis buffer with 1% phenylmethylsulfonyl fluoride (PMSF) using an electrical homogenizer at 5000 Hz for 30 s. After centrifugation at 10,000*g*, 4 °C for 5 min, the supernatant was collected, and the protein level was detected using a BCA protein assay kit with bovine serum albumin as a standard (Bio-Swamp, China). The supernatant was prepared into samples (5 μg/μl) with loading buffer containing 250 mM Tris (pH 6.8), 10 mM DTT, 10% SDS, 0.5% bromophenol blue and 50% glycerol, according to their determined protein level. The samples were loaded onto a 10% polyacrylamide gel. Proteins were then transferred to PVDF membranes (Biogood, China). The membrane was blocked with 5% non-fat, dry milk in TBST for 60 min at room temperature to lower the background. The membranes were then incubated with polyclonal rabbit antibodies against AQP4 (Abcam, UK) and Caspase3 (Cell Signaling Technology, USA) overnight at 4 °C. The membranes were subsequently incubated with peroxidase-conjugated goat anti-rabbit IgG (Biogood, China) and exposed to an ECL kit (Bio-Swamp, China). The membranes were analyzed using an ECL scanner (Bio-Rad, USA).

## IMAGE PROCESSING AND STATISTICAL ANALYSIS

The DWI images were fitted into ADC maps and analyzed by Statistical Parametric Mapping 12 (SPM12, http://www.fil.ion.ucl.ac.uk/spm/). Firstly, all b0 images (b value = 0) from each animal were normalized spatially into a selected preferable b0 image, and average all normalized-b0 images to get a b0 template. Secondly, each b0 image was spatially into the b0 template to get deformation field information, which was used to normalize each ADC map into the b0 template. Thirdly, all modulated ADC maps were smoothed with a 0.2-mm full width, at half maximum (FWHM) Gaussian kernel. The intergroup differences of the sham-acute hypoglycemia group, the acute hypoglycemia- recovery group and the sham-recovery group in the ADC map were assessed with voxel-wise two-sample t-tests. The threshold for statistical significance was set to $P < .01$, FDR corrected. The minimum cluster size was set to 20.

The Sharpiro-Wilk test and Levene's test were performed to ensure the normal distribution and equal variance of the

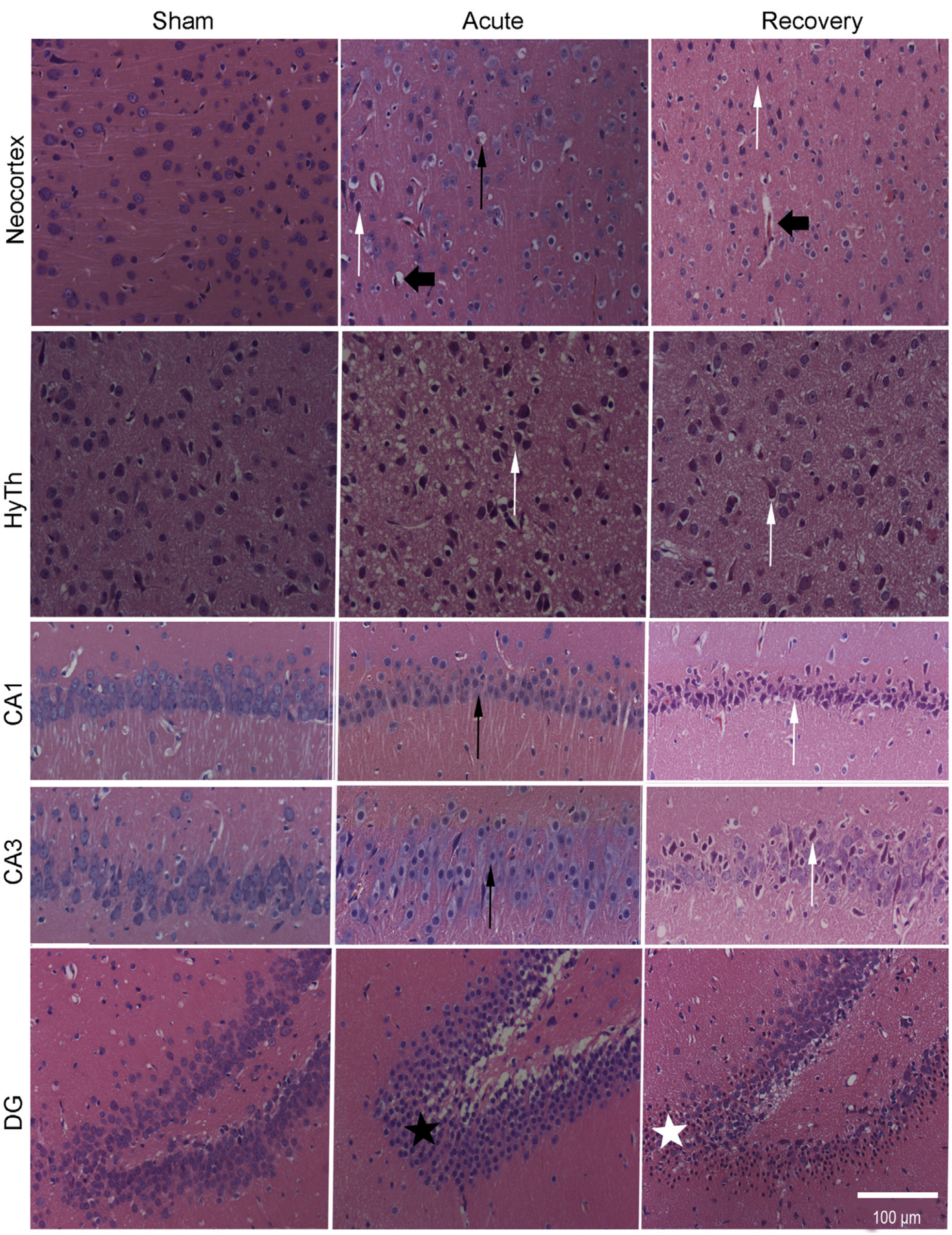


**Fig. 2.** Neuropathologic changes observed with H&E staining. Neocortex and subcortex: widened Virchow-Robin space, swollen neurons and several degenerated neurons in the acute hypoglycemia group; shrunken and necrotic neurons in the superficial layers in the recovery group. Hypothalamus: degenerated neurons in the acute hypoglycemia group; several red neurons in the recovery group. Hippocampus: swollen neurons in CA1 and CA3 and pyknotic granule neurons in DG of the acute hypoglycemia group; missing neurons in the dentate crest and delayed selective neuronal death in CA1 and CA3 in the recovery group. Black arrows (thin): swollen neurons; black arrows (thick): widened Virchow-Robin space; white arrows: degenerated, necrotic and red neurons; black five-pointed star: pyknotic granule neurons; white five-pointed star: missing granule neurons.

data in advance. Then, a one-way ANOVA test was conducted to analyze the neuronal size and the expression of the proteins (AQP4 and Caspase3) among the three groups. In cases where the ANOVA test demonstrated significant results, post-hoc LSD t-tests were subsequently performed. Data was analyzed using IBM SPSS Statistics 20.0 (https://www.ibm.com/products/software). The threshold for statistical significance was set to $P$ < .05.

## RESULTS

### DWI findings

As is illustrated in Fig. 1, the acute hypoglycemia group showed increased ADC in the DG and hypothalamus when compared to the sham group. The increased ADC region of the hypothalamus involved dorso medial hypothalamus (DMH) and at lateral hypothalamus (LH). The acute hypoglycemia group also demonstrated a decreased ADC in the somatosensory cortex (SSc), subcortex and striatum (Str) compared to the sham group. After glucose reperfusion and 7 days of recovery, most of the hypoglycemia-induced, abnormal ADC, returned to normal except for that in the LH of the hypothalamus and the dentate gyrus (DG), where increased ADC levels persisted. In the recovery group, the posterior SSc and its subcortex demonstrated an increased ADC when compared to the sham group.

### Neuropathologic findings

*Neuropathologic changes by H&E staining (Figs. 2 and 3)*

Cortex and subcortex: A widened Virchow-Robin space, swollen neurons and neurons with minor degeneration could be observed in the acute hypoglycemia group. In the recovery group, degenerated neurons could be found in the superficial layers. The neuronal volume was shrunken. One-way ANOVA test revealed significant size change among the three groups ($F(2,12) = 71.891$, $P$ < .001). Post-hoc comparison demonstrated that the neuron size of the acute hypoglycemia group was significantly larger when compared to the sham group ($P$ = .002). As for the recovery group, the pathologic changes in the cortex were not as severe as the acute hypoglycemia group, but the average size of the neurons was shrunken when compared to the sham group ($P$ < .001). The white matter of

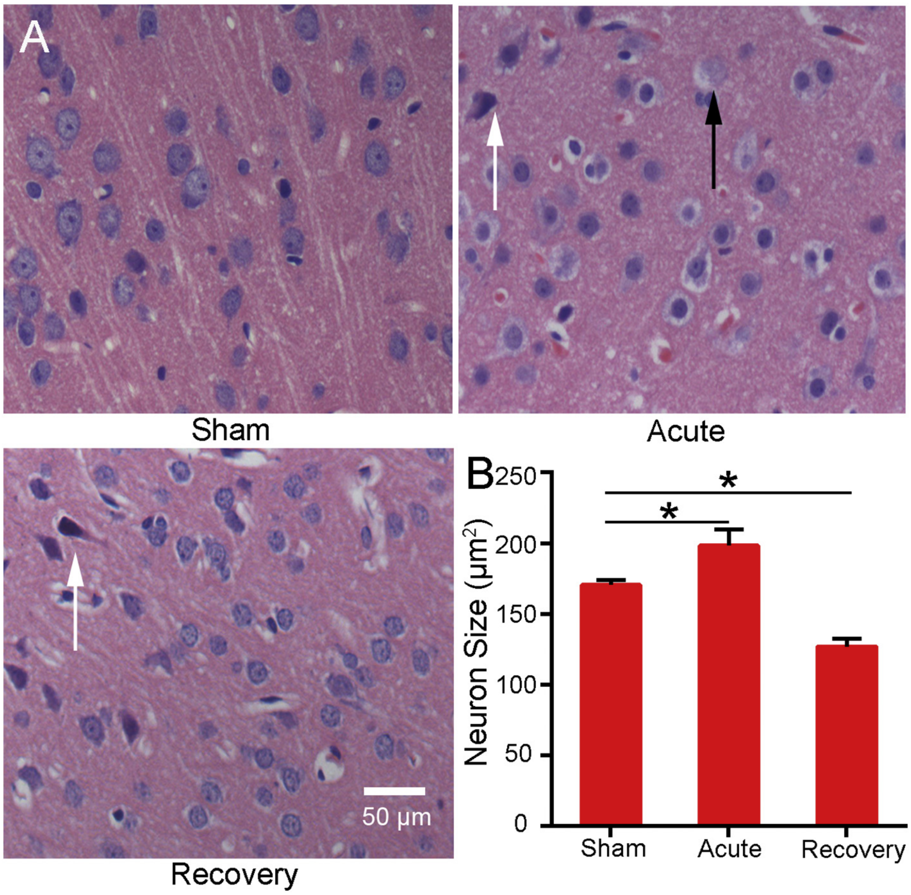


**Fig. 3.** The high power microscopy of the cortical neuropathologic changes observed with H&E staining. A. Swollen neurons in the acute hypoglycemia group and shrunken neurons in the recovery group. B. Larger neuron size in the acute hypoglycemia group when compared to the sham group (*$p < 0.05$). Shrunken neuron size in the recovery group when compared to the sham group (*$p < 0.05$). Black arrows (thin): swollen neurons; black arrows (thick): widened Virchow-Robin space; white arrows: degenerated, necrotic and red neurons; black five-pointed star: pyknotic granule neurons; white five-pointed star: missing granule neurons.

the subcortex did not demonstrate conspicuous pathologic changes, except for unspecific edema among axons of neurons in both the acute hyperglycemia and recovery groups.

Hypothalamus: In the acute hyperglycemia group, the main pathologic change in the hypothalamus was degenerated neurons. The neurons appeared hyperchromatic with pyknotic nuclei. In the recovery group, after 7 days of glucose reperfusion, the pathologic changes were alleviated but several red neurons were still observed.

Hippocampus: in the acute hypoglycemia group, swollen neurons were observed in cornuammonis region 1 (CA1) and cornuammonis region 3 (CA3) and granule neurons in the DG became pyknotic. In the recovery group, several neurons in the dentate crest were missing and we found delayed, selective neuronal death in CA1 and CA3.

## Neuropathologic changes by silver staining (Fig. 4)

Cortex: Swollen axons could be observed in both the acute hypoglycemia and recovery groups, but the changes in the former group were more severe.

Subcortex: Except for edema, no conspicuous neuropathologic changes, such as Wallerian degeneration and retraction balls, were observed in the acute hyperglycemia or recovery groups. The edema in the acute hypoglycemia group was more severe than that in the recovery group.

Hippocampus: The axons in the acute hypoglycemia group did not show obvious pathological changes when compared to the sham group. The axons were intact and were normal in appearance. In the recovery group, after 7 days of glucose reperfusion, the axons became disorganized and disrupted.

Hypothalamus: In the acute hypoglycemic group, the brain tissue of the hypothalamus was loose and a few swollen axons were observed. In the recovery group, a few axons showed a mild wavy appearance.

## The expression of AQP4

The western blot analysis revealed an approximately 48 kDa band for AQP4. The relative expression levels of AQP4 in the cortex, hypothalamus and hippocampi were illustrated in Fig. 5. In the cortex, a one-way ANOVA test demonstrated a significant result among the three groups ($F(2,12) = 9.288$, $P = .004$). Post-hoc comparison showed that the expression of AQP4 in the acute hypoglycemia group was significantly lower than both the sham ($P = .003$) and the recovery group ($P = .003$). In the hippocampus or hypothalamus, no significant difference in AQP4 expression was observed when applying the one-way ANOVA test ($F(2,12) = 1.934$, $P = .187$; $F(2,12) = 2.062$, $P = .170$).

## The expression of Caspase3

Western blot analysis revealed an approximately 35 kDa band for Caspase3. The relative expression levels of Caspase3 in the cortex, hypothalamus and hippocampi were illustrated in Fig. 6. Using the one-way ANOVA test, we did not find a statistically significant change in Caspase3 expression in the cortex ($F(2,12) = 1.723$, $P = .220$). In the hippocampus, the one-way ANOVA test demonstrated significant differences in the Caspase3 expression between the three groups ($F(2,12) = 16.903$, $P < .001$). Post-hoc comparison revealed a significant increase in Caspase3 expression in the recovery group when

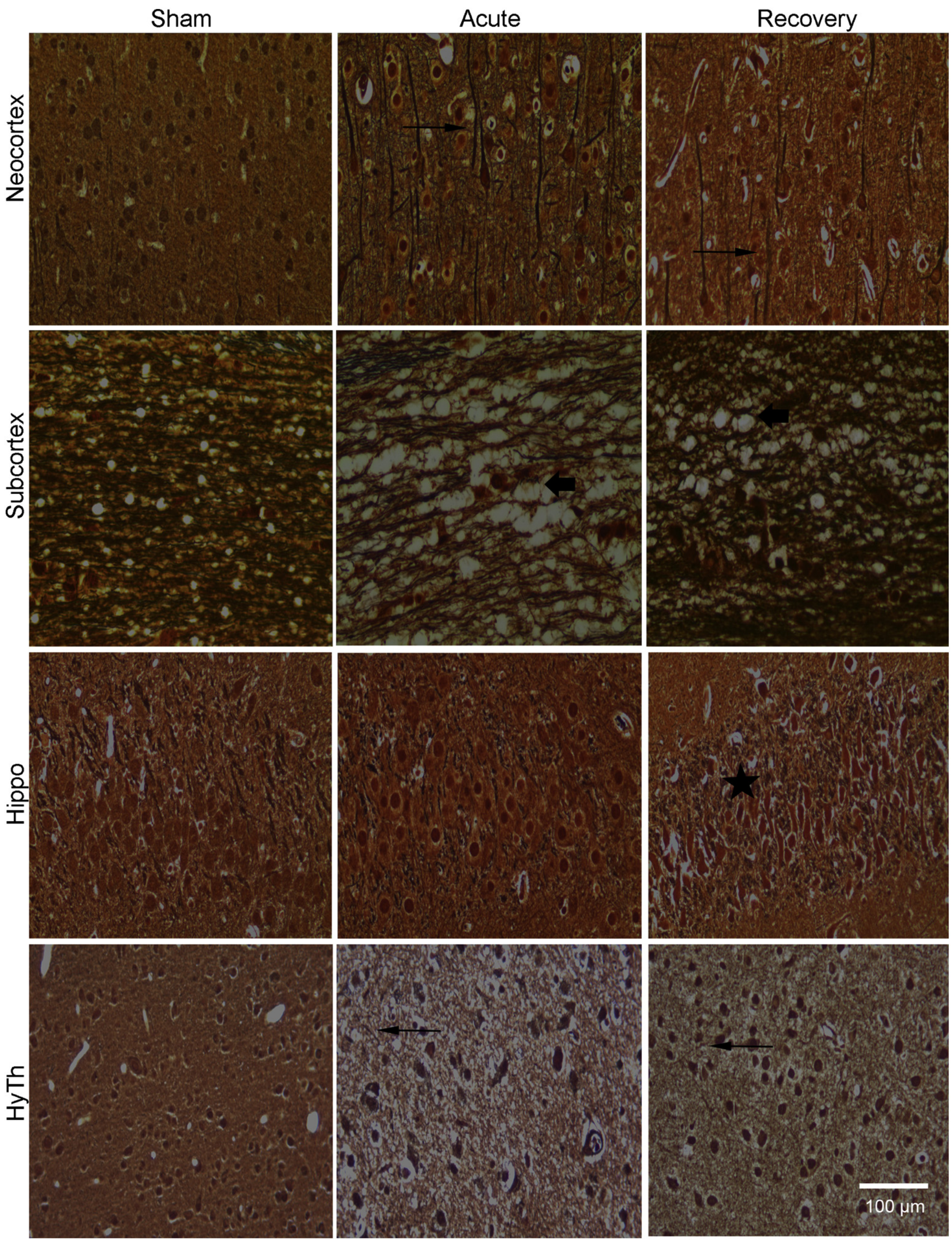


**Fig. 4.** Neuropathologic changes observed with silver staining. Neocortex: swollen axons in the cortex of both of the acute hypoglycemia and the recovery groups. Subcortex: edema among axons in both the acute hypoglycemia and the recovery groups. Hippocampus: normal axons in the acute hypoglycemia group; became disorganized and disrupted axons in the recovery group. Hypothalamus: loose brain tissue and a few swollen axons in the acute hypoglycemia group; mild wavy axons in the recovery group. Black arrows (thin): swollen axons; black arrows (thick): edematous white matter; white arrows: wavy neurons; black five-pointed star: disrupted axons.

compared to the sham ($P$ < .0001) and acute hypoglycemia groups ($P$ < .0001). One-way ANOVA demonstrated a significant result in the hypothalamus (F(2,12) = 4.038, $P$ = .046). Post-hoc comparison found that the Caspase3 expression in the hypothalamus was increased in the acute hypoglycemia group when compared to the sham group ($P$ = .015).

All the results above are summarized in Table 1.

## DISCUSSION

The results demonstrated asymmetric ADC-abnormal regions in the cerebral cortex, the subcortex, the hippocampus, and the hypothalamus. Water diffusion changes in the brain are generally suggestive of damage. Thus, these locations were closely focused on during the subsequent examinations. The following neuropathologic findings revealed a broader distribution than the ADC map, with complex pathologic changes. The major pathologic change observed in the present study was edema, which is considered to be one of the main indications of changes in water diffusion. Deng et al. revealed that brain edema was triggered immediately after severe hypoglycemia and continued to progress even after recovery (Deng et al., 2014). According to previous research, brain edema could be classified into vasogenic and cytotoxic types (Klatzo, 1967). It is well known that decreased ADC is suggestive of cytotoxic edema in the practice of clinical medicine, but the diagnosis depends on the signal of the T2 weigh and other sequences (Mintorovitch et al., 1991; Latour et al., 1994; Schlaug et al., 1997). Clinical trials using DWI have revealed reversible cytotoxic edema in hypoglycemic encephalopathy in the cerebral cortex, basal ganglia (BG), hippocampus, splenium and internal capsule (IC) (Boeve et al., 1995; Maekawa et al., 2006; Yanagawa et al., 2006; Kang et al., 2010; Johkura et al., 2012; Ohshita et al., 2015). The cytotoxic edema mainly invades the parenchyma, with an intact BBB and increase in cell size (Klatzo, 1967). The neuropathologic findings of ubiquitous swollen neurons was in accordance with the pathologic changes expected in cytotoxic edema (Klatzo, 1967). The locations mentioned in our study were mostly consistent with the previous research. The mechanism of cytotoxic edema is

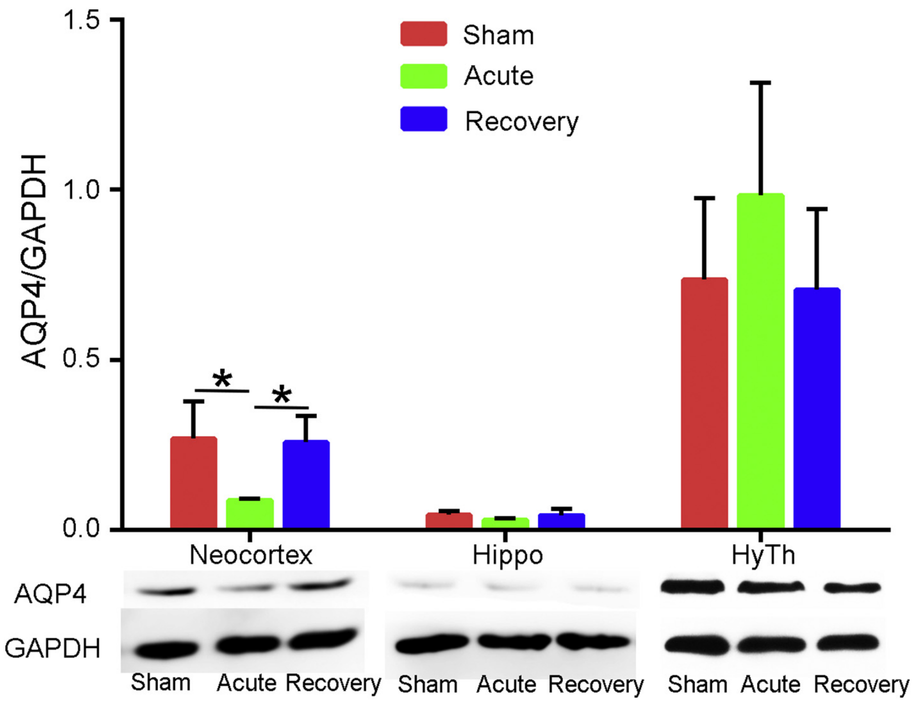


**Fig. 5.** Western blot on AQP4. In the cortex, one-way ANOVA and post-hoc comparisons demonstrated that the lowest expression was in the acute hypoglycemia group (*$p$ < 0.05). In the hippocampus and hypothalamus, no statistically significant differences in AQP4 expression was observed between the three groups.

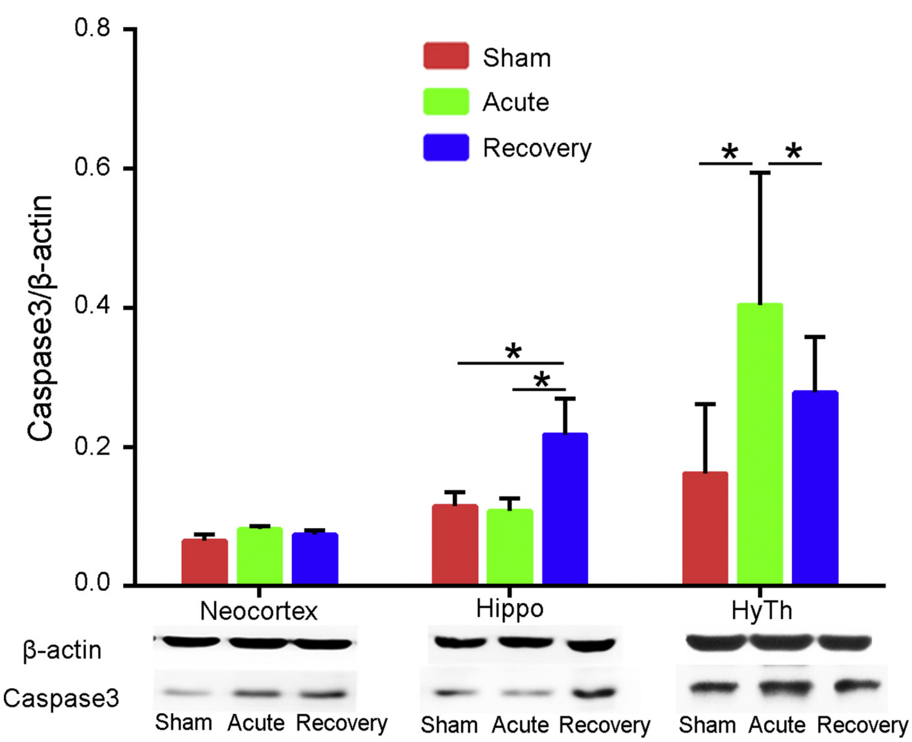


**Fig. 6.** Western blot on Caspase3. In the hippocampus, one-way ANOVA and post-hoc comparisons demonstrated that the Caspase3 expression of the recovery group was increased when compared to both the sham and acute hypoglycemia groups (*$p$ < 0.05). In the hypothalamus, the expression of Caspase3 peaked in the acute hypoglycemia group (*$p$ < 0.05). We did not find a statistically significant change for Caspase3 expression in the cerebral cortex.

related to the disturbance of cell osmoregulation induced by the dysfunction of the ATP-dependent sodium pump (Klatzo, 1967). Hypoglycemia is an event of metabolic disorder and will inactivate the channels on the cell membrane, which contribute to the restriction of fluid flow (Lopez-Gambero et al., 2019). Vasogenic edema may partly account for the increased ADC regions in this research (Mintorovitch et al., 1991). The primary injury associated with vasogenic edema is to the BBB, which will enhance the vascular permeability and increase the water diffusion (Klatzo, 1967). Thus, the Virchow-Robin space was widened under these circumstances. The breakdown of the BBB in severe hypoglycemia has been validated and thought to be the primary pathogenesis of vasogenic edema (Deng et al., 2014). Vasogenic edema enhances the extracellular volume and decreases the restriction on water flow, which leads to an ADC increase (Pierpaoli et al., 1993). As we know, ADC reflects a mixture of pathologic events (Schlaug et al., 1997). Both types of edema existed in the present study and the regional ADC changes may depend on the predominant type of edema. In our study, regions of SSc and Str demonstrated decreased ADC in the acute stage of hypoglycemia, which suggested that cytotoxic edema was more prominent than vasogenic edema during this stage. We also observed an interesting phenomenon, whereby some normal regions of the posterior SSc in the acute hypoglycemic stage, later showed an increased ADC after 7 days recovery. This has also been reported in previous research where the ADC increased 7 days post-cytotoxic insult (Schlaug et al., 1997). It is thought that the normal ADC observed during the acute hypoglycemic stage, was due to a balance between cytotoxic and vasogenic edema. After recovery, cytotoxic edema was alleviated but vasogenic edema persisted, which gave rise to the increased ADC. Our neuropathologic findings

**Table 1.** Findings of DWI imaging, neuropathologic examination and Western blot.

| Examination | Cerebral Cortex and Subcortex | | Hypothalami | | Hippocampi | |
|---|---|---|---|---|---|---|
| | Acute | Recovery | Acute | Recovery | Acute | Recovery |
| DWI | Decreased ADC in anterior SSc and subcortex | Increased ADC in posterior SSc and subcortex | Increased ADC in DMH and LH | Increased ADC in LH | Increased ADC in DG | Increased ADC in DG |
| H&E Staining | Widened Virchow-Robin space, degenerated and swollen neurons | Widened Virchow-Robin space, shrunk and degenerated neurons in the superficial layers | Degenerated neurons | Several red neurons | Swollen neurons and pyknotic granule neurons in the DG | Missing neurons in the dentate crest and delayed neuronal death in CA1 and CA3 |
| Silver Stain Impregnation | Severely swollen axons in the cortex and edematous white matter in the subcortex | Swollen axons in the cortex and mildly edematous white matter in the subcortex | Loose brain tissue and a few swollen axons | Mildly wavy axons | Minor pathological changes | Disorganized and disrupted axons |
| Western Blot on AQP4 in the Cortex | Upregulated AQP4 expression | No statistical difference | No statistical difference | No statistical difference | No statistical difference | No statistical difference |
| Western Blot on Caspase3 in the Cortex | No statistical difference | No statistical difference | Upregulated Caspase3 expression | No statistical difference | No statistical difference | Upregulated Caspase3 expression |

strengthen this theory. The insignificant changes in expression of Caspase3 in the cerebral cortex indicated that cytotoxic edema may exist without neuronal apoptosis in severe hypoglycemia. As for the subcortex, though ADC changes occurred in the subcortical SSc, with the exception of edema, no conspicuous pathologic changes, such as Wallerian degeneration and retraction balls, were observed. We considered that the decreased ADC in the acute hypoglycemic stage was ascribed to the severe metabolic disorder, while the increased ADC after 7 days recovery was due to the persistent vasogenic edema.

In addition to edema, we should take into consideration functional factors that explain the increased ADC in the hypothalamus and hippocampus. The increased ADC in the hypothalamus may be partly ascribed to some functional changes since the hypothalamus exercises central control in energy homeostasis and the neurons in the hypothalamus are sensitive to changes in glucose levels (Lopez-Gambero et al., 2019). The lateral hypothalamus (LH) is validated to be the orexigenic centre (Lopez-Gambero et al., 2019) and the increased ADC regions in our results involved the LH. Additionally, the hypothalamus contains glucose-excited neurons which are pivotal to the initiation of the counter regulatory response to hypoglycemia (Burdakov et al., 2005; Melnick et al., 2011; Sprague and Arbeláez, 2011). Therefore, we speculated that the increased water diffusion may be due to hypothalamic response to glucose deprivation. Previous studies rarely mentioned hypothalamic damage due to hypoglycemia. As we know, neuronal necrosis may also increase the ADC value. The pathologic changes in the hypothalamus involved moderate neuronal and axonal degeneration. Western blot results demonstrated that hypothalamic neuronal apoptosis was initiated in the acute hypoglycemia stage and was alleviated after recovery. These findings suggested that the hypothalamus is a specific region which is vulnerable to hypoglycemia.

Hippocampal damage induced by hypoglycemia consisted of a mixture of metabolic disorder as well as excitotoxic injury. The swollen neurons observed in CA1 and CA3 demonstrated the presence of metabolic disorder. In the increased ADC regions of the hippocampus, we observed severe neuronal damage in the DG in both the acute hypoglycemia and recovery group. In addition, we observed delayed neuronal death in Ammon's horn in the recovery group. The upregulation of Caspase3 expression in the Western blot of the hippocampi suggested a delayed neuronal damage. The DG damage and delayed neuronal death should be ascribed to the excitotoxic brain injury described in previous research (Auer et al., 1985a). Neural chemistry studies demonstrated that hypoglycemia causes a flood of excitotoxic amino acids (glutamate/aspartate) into the extracellular space of the brain (Norberg and Siesio, 1976; Sandberg et al., 1986; Auer, 2004). It is known that the hippocampus controls cognition and contains a high density of NMDA receptors (Monaghan et al., 1983). The fluid exchange will be intensified with the activation of these NMDA receptors, which in turn may increase the water diffusion. The present research demonstrated that the effect of excitotoxic injury may play a key role in the ADC changes when compared with the effects of metabolic disorder in the hippocampus. The extracellular glutamate/aspartate indiscriminately kills hippocampal neurons and this was a possible agent contributing to the hippocampal necrosis (Auer et al., 1985b; Won et al., 2012; Kho et al., 2017). In severe hypoglycemia, the flood of extracellular glutamate/aspartate will activate the NMDA receptors and trigger an influx of calcium (Páramo et al., 2013), which initiates a sequential release of nitric oxide, zinc and superoxide in the brain (Suh et al., 2008). This will damage the DNA of neurons and lead to neuronal necrosis (McGowan et al., 2006). The neuronal degeneration in the hippocampus could be alleviated through the blocking of the NMDA receptors in a hypoglycemic vivo model (Paramo et al., 2010). A recent study conducted by Kho et al. (Kho et al., 2017) found that Nacetyl-l-cysteine (NAC), a broad spectrum antagonist of oxidation, could effectively prevent delayed neuronal death.

The hippocampal injury mentioned above is thought to be one of the specific mechanisms of damage triggered by severe hypoglycemia. This has forensic value in distinguishing between hypoglycemia and ischemia. Canonical neuropathologic studies revealed the difference between hypoglycemic and ischemic injury in the brain (Auer, 2004). Ischemic brain damage only leads to global damage inside the ischemic border and rarely gives rise to DG necrosis (Auer et al., 1984; Auer and Siesjo, 1988). In addition, severe hypoglycemia tends to cause mild injury in the superficial layers of the cortex and seldom injures the cerebellum and brainstem (Auer et al., 1984; Auer and Siesjo, 1988).

AQP4 is a pivotal protein in edematous conditions (Assentoft et al., 2015), however the real function of AQP4 in edema is still elusive. Previous research revealed a decrease in edema formation and animal mortality with AQP4 deletion in ischemia, water intoxication, acute liver failure and meningitis (Manley et al., 2000; Papadopoulos and Verkman, 2005; Rama Rao et al., 2014) whereas there was an increase when overexpressed in water intoxication (Yang et al., 2008). Other conditions such as brain tumors, hypothermic injury and staphylococcal brain abscess can intensify edema formation and animal mortality in AQP4$^{-/-}$ mice (Papadopoulos et al., 2004; Bloch et al., 2005). According to previous studies, the regulatory pattern of AQP4 expression is disputable in ischemia (Assentoft et al., 2015). Amiry-Moghaddam et al. found that there was downregulation of AQP4 expression in the ischemic core and penumbra after middle cerebral artery occlusion (MACO) (Amiry-Moghaddam et al., 2003). Frydenlund et al. also demonstrated downregulated perivascular AQP4 at the ischemic striatum and cortex while it was upregulated in the border zone after MACO (Frydenlund et al., 2006). Other research (Kleindienst et al., 2006; Okuno et al., 2008; Higashida et al., 2011) showed entirely opposite results, namely that the AQP4 expression was increased after ischemia. While, Kim et al. (Kim and Jung, 2011) found no difference in AQP4 expression in the ischemic regions. Deng et al. observed that AQP4 was upregulated 24 h after the recovery of hypoglycemia and speculated that AQP4 is a protective protein in hypoglycemia (Deng et al., 2014). The research on AQP4 in severe hypoglycemia is limited, thus we compared the AQP4 expression with ADC in the cerebral cortex, hippocampus and hypothalamus. The only finding was that the downregulation of AQP4 in the cerebral cortex of the acute hypoglycemia group, was parallel with the decreased ADC level. In the hypothalamus, we observed that AQP4 was upregulated without statistically significant differences in the acute hypoglycemia group. The results suggested that AQP4 was associated with the water diffusion as well as the formation of cytotoxic edema in acute hypoglycemia.

We must disclose some limitations in our present study. The ADC map from the DWI imaging is for the general distribution of water diffusion. It can partly mirror the neuropathologic changes but not in their entirety. In addition, the study was conducted using rat models and the situation would be more complicated in human studies. The differences in the laterality of the ADC imaging between hemispheres cannot be elucidated in this research and is a flaw in this study which will be evaluated in future research.

In conclusion, our work revealed that hypoglycemia significantly influenced the water diffusion of the brain. Severe hypoglycemia caused cytotoxic, as well as vasogenic edema in the acute stages. After recovery, cytotoxic edema was alleviated while vasogenic edema persisted. The decrease of AQP4 was associated with the formation of cytotoxic edema in acute hypoglycemia. Hypoglycemia primarily tends to damage the cerebral cortex, hippocampus and hypothalamus and may result in permanent injury to the brain.

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